

Physical Simulation of Marsupial UAV-UGV Systems Connected by a Variable-Length Hanging Tether

J. E. Maese¹, L. Merino¹ and F. Caballero¹

Abstract— This paper presents a simulation framework capable of modeling the dynamics of a hanging tether of adjustable length that connects a UAV to a UGV. The model incorporates the interaction between the UAV, UGV, and a winch, allowing for dynamic tether adjustments based on the relative motion of the robots. The accuracy and reliability of the simulator are assessed through extensive experiments, including comparisons with real-world experiments, to evaluate its ability to reproduce the complex tether dynamics observed in physical deployments. The results demonstrate that the simulation closely aligns with real-world behavior, particularly in constrained environments where tether effects are significant. This work provides a validated tool for studying tethered robotic systems, offering valuable insights into their motion dynamics and control strategies.

I. INTRODUCTION

Marsupial systems, like those where unmanned aerial vehicles (UAVs) are deployed from unmanned ground vehicles (UGVs), offer a powerful combination of aerial and terrestrial capabilities for applications such as search and rescue, infrastructure inspection, and military missions. Tethered configurations extend the operational time of the UAV by providing continuous power from the UGV, significantly enhancing the range and flexibility of the mission. However, tethers restrict maneuverability, which requires advanced planning algorithms and an appropriate simulation framework to address coordination challenges.

A trajectory planning method with an adjustable tether was proposed in [1], enhancing the flexibility of the UAV. This contrasts with early studies on tethered UAV-UGV systems, such as [2], which assumed a taut tether and limited the maneuverability of UAVs. These systems have been successfully applied in real-world scenarios such as the DARPA Subterranean Challenge, where the CSIRO team [3] used them for UAV-UGV collaboration in hazardous environments. Similarly, tethered helicopter landings have been explored in [4], demonstrating their relevance in UAV-UGV coordination.

Despite these advances in UAV-UGV applications, tether modeling and control have been studied independently of simulation frameworks. Finite element models have been



Fig. 1: 3D LiDAR reconstruction of the theatre in Gazebo where the experiment was conducted (left) and the real-world experiment used as a reference (right).

used to capture the mechanical properties of tethers, although without integration with UAV-UGV kinematics [5]. Other approaches treated the tether as an additional degree of freedom, improving length control yet without accounting for slack behavior or environmental interactions [6]. Underwater robotics research has also contributed relevant insights: a catenary-based visual servoing scheme was proposed for tether shape regulation in underwater vehicles [7]; the catenary shape control concept transfers to aerial systems, where aerodynamic drag is small relative to gravitational load at operational speeds. Path planning has likewise been refined by incorporating tether constraints, reducing risks during physical testing [8]. While these works advance individual aspects of tether modeling and control, they do not provide an integrated simulation framework for real-time UAV-UGV coordination.

Common robotic simulation platforms, such as Gazebo, have been employed for UAV-UGV studies, yet typically rely on simplified tether models that do not capture real-time dynamic interactions. Packages such as `sitl_gazebo` [9] extend UAV simulation capabilities in Gazebo but lack flexible tether management for complex scenarios. A catenary model has also been used to compute tether configurations for trajectory optimization, although without simulating real-time tether physics [1]. There has thus been a lack of comprehensive simulation frameworks that fully integrate real-time UAV-UGV control with a physically accurate, variable-length tether model.

This paper presents a physics-based simulator capable of modeling a hanging tether with variable length (Fig. 1). This simulator fully integrates tether physics with UAV and UGV dynamics within a ROS 2-based Gazebo environment, enabling researchers to test and optimize control strategies in a risk-free setting. The simulation framework is validated

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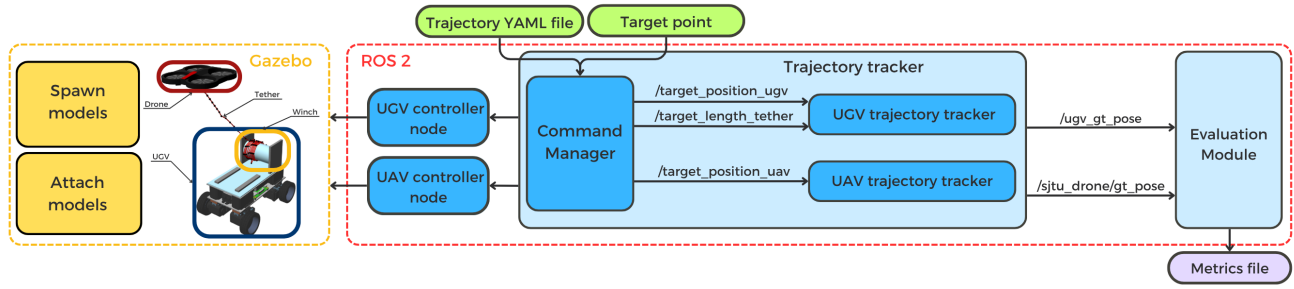


Fig. 2: Diagram of the Marsupial Simulator. The core modules of simulator are in blue. The base robotics simulator modules are depicted in light yellow color. Green boxes indicate the input data information. Output evaluation metrics are indicated in light purple color.

through extensive experiments, including comparisons with real-world trajectories to assess its accuracy in replicating complex tether dynamics. By providing a detailed and validated simulation platform, this work contributes to the development of more robust and efficient UAV-UGV tethered systems.

II. MARSUPIAL SYSTEM SIMULATOR USING GAZEBO

The marsupial UAV-UGV simulator, built with ROS 2 and Gazebo, models tethered robot dynamics and supports both manual and autonomous operation (Fig. 2). The main components are:

- **Model Initialization:** The simulation spawns the UGV, UAV, and a coiled tether in Gazebo, with the UAV positioned on a platform on the UGV. ROS 2 modules handle system initialization and execution.
- **Trajectory Tracking:** The system follows predefined YAML waypoints with reference tether lengths or real-time ROS messages. The UAV, UGV, and winch dynamically adjust to maintain proper tether slack, supporting custom trajectory algorithms.
- **Controllers:** Dedicated controllers manage UAV and UGV movement, with the UGV also handling tether adjustments. The control framework ensures realistic behavior and can be modified for different dynamics.
- **Evaluation and Data Recording:** The system records UAV/UGV poses, tether length variations, and key performance metrics, storing data for post-analysis of trajectory accuracy, tether behavior, and system stability.

The simulator is modular; full documentation and open-source code are available in the Supplementary Materials.

A. Models

The simulation environment integrates established Gazebo models for the UAV [10] and UGV [11] with a custom-developed winch and a variable-length tether adapted from [9], all unified under a novel ROS 2 architecture that coordinates their interaction in real time (Fig. 2, Gazebo section).

The UAV model [10] is a quadrotor equipped with sensors such as cameras, IMU, sonar, LiDAR, and GPS for navigation and control. It supports both position and velocity

control modes and provides ground-truth feedback for precise evaluation. To integrate into the marsupial system, its thrust along the Z-axis was increased to compensate for the tether's weight; the tether force itself, including horizontal components, is applied to the UAV body directly by Gazebo's physics engine through the tether attachment joint.

The UGV model [11] is based on a holonomic platform, allowing omnidirectional movement. This enhances maneuverability in constrained environments and facilitates UAV-UGV coordination while managing the tether. Parameters such as wheel effort limits, steering joint dynamics, and winch joint damping were tuned through simulation trials, starting from the default values of the base platform, until three criteria were met: stable UGV motion at the operating speed of 0.5 m/s without oscillations, smooth winch spooling without backlash, and robustness to the lateral forces exerted by the tether during UAV maneuvers.

The winch, custom-developed and integrated into the UGV, dynamically adjusts tether length based on UAV-UGV relative positions or predefined references. It includes a rotating cylinder to maintain proper slack and a stable platform for UAV takeoff. The winch is controlled by the UGV's module, ensuring synchronized tether management during both manual and autonomous operation.

The tether model discretizes the cable into $N=125$ rigid elements of length $l=0.15$ m, radius $r=4$ mm, and mass $m_i=\rho_l l$ with $\rho_l=0.01$ kg/m. Elements are connected by universal joints whose restoring torque follows a spring-damper law:

$$\tau_j = -d\dot{\theta}_j - k_s \theta_j, \quad (1)$$

with damping $d=0.1$ N·m·s/rad and stiffness $k_s=0.01$ N·m/rad. Aerodynamic resistance on each element is modelled as $F_{d,i} = \frac{1}{2}\rho_{\text{air}}\|v_i\|^2 C_d A_i$, with $\rho_{\text{air}}=1.204$ kg/m³, $C_d=1.25$, and projected area $A_i=2rl$.

Physical parameters (r , ρ_l) are representative of lightweight cables used in tethered UAV systems. The drag coefficient corresponds to the standard value for a long cylinder in crossflow. Joint damping and stiffness were adjusted iteratively in simulation to ensure stable catenary dynamics without oscillatory artifacts. The number of elements N is the dominant parameter governing the accuracy-performance trade-off: the RTF decreases from 0.99 ($N=100$) to 0.19

TABLE I: Simulation Metrics and Graphs

Metrics		
Metric	Description	Units
sim.time	Simulation duration	s
targets	Number of target points	-
dist.ugv	Distance traveled by UGV	m
dist.uav	Distance traveled by UAV	m
tether.released	Tether released	m
tether.collected	Tether collected	m
cat.error	Median catenary error	m
Graphs		
Graph	Description	Units
pos_3D	3D position visualization	m
uav_pos	UAV position on axes	m
ugv_pos	UGV position on axes	m
uav_vel	UAV velocity	m/s
ugv_vel	UGV velocity	m/s
tether.len	Released length of the tether	m
min.dist	Minimum distance to obstacles	m

($N=700$) on the reference hardware (Fig. 5), and $N=125$ was selected as the value that best balances computational efficiency and discretization accuracy.

III. METRICS FOR SIMULATION EVALUATION

The simulator records UAV and UGV velocities, positions, and tether length variations, capturing key aspects of movement and tether dynamics (Table I). A key goal of these metrics is to validate the realism of the simulated physical dynamics, particularly those governing the tether's behavior.

The total distance traveled by the UGV and UAV is computed as the sum of Euclidean distances between successive waypoints. The tether length adjustments are estimated by summing the positive and negative increments over time:

$$L_{released} = \sum_{i=1}^{n-1} \max(0, L_{i+1} - L_i) \quad (2)$$

$$L_{collected} = \sum_{i=1}^{n-1} \max(0, L_i - L_{i+1}) \quad (3)$$

Additionally, the simulator's accuracy is evaluated by comparing the tether shape to an ideal catenary curve, analyzing the impact of element discretization on realism. The catenary error `cat_error` is computed at each time step by fitting a theoretical catenary to the two deployed tether endpoints given the actual deployed length, then measuring the mean nearest-neighbour distance from each simulated element to that curve; the reported value is the median over all time steps. These metrics also provide insights into UAV-UGV navigation and tether behavior during deployment and retraction, ensuring consistency with expected physical properties such as rigidity, elasticity, and dynamic interactions.

All simulation data is automatically recorded in structured CSV files (UAV/UGV kinematics, winch state, and tether element positions), enabling post-analysis and visualization of every metric in Table I.

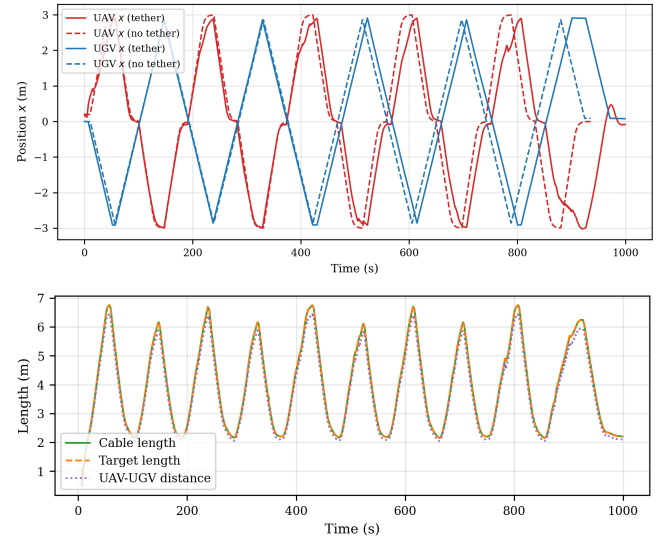


Fig. 3: Opposite Direction Coordination results. Top: x -position of UAV and UGV with tether (solid) and without tether (dashed). Bottom: cable length, target length, and UAV-UGV distance.

IV. VALIDATION EXPERIMENTS

To validate the simulator, we designed scenarios assessing stability, tether dynamics, winch operation, and UAV-UGV coordination. These are grouped into Simulated Scenarios, evaluating general functionality, and Real Scenarios, replicating real-world experiments. A video summary showcasing the simulator's capabilities is provided in the Supplementary Materials.

A. Simulated Scenarios

The simulated scenarios evaluate the simulator's ability to handle UAV-UGV dynamics under different conditions. Three key tests were designed: (1) Vertical Stability Assessment, where the UAV ascends and descends while the UGV remains stationary, assessing the winch's responsiveness; (2) Horizontal Mobility Assessment, where the UGV moves while the UAV hovers, analyzing tether slack management; and (3) Opposite Direction Coordination, where both robots move in opposite directions, challenging system coordination.

The UAV and UGV successfully completed their trajectories across all simulated scenarios. Fig. 3 presents detailed results for the Opposite Direction Coordination scenario. The top panel compares the x -position trajectories with and without the tether: while both cases reach all reference waypoints, the tethered system does so with a delay of approximately 6.5 s, attributable to the additional dynamic load introduced by the cable on both vehicles. This delay is consistent with the added inertial and drag loading introduced by the tether model, confirming its ability to capture the mechanical coupling between the robots. The bottom panel shows that the tether controller closely tracks the target length reference

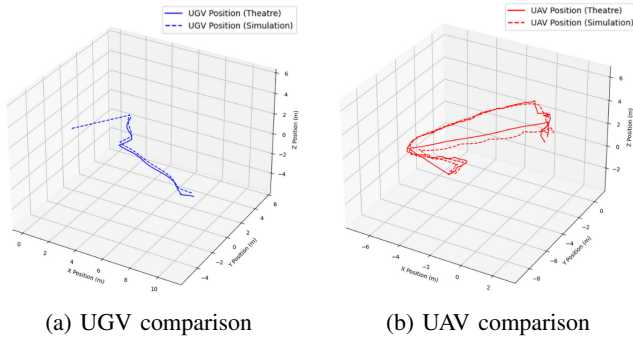


Fig. 4: 3D position comparison: UGV and UAV real theatre experiment vs simulation scenario.

TABLE II: Simulation results of the real experiment, including simulation time (s), number of targets, UAV/UGV distance traveled (m), and tether released/collected (m)

Scenario	Sim. time	Targets	Dist. UAV	Dist. UGV	Tether rel.	Tether col.
RTTA	628	100	33.92	16.45	17.91	7.54
PTR	609	100	33.96	16.41	19.57	8.40

throughout the experiment, maintaining appropriate slack without excessive tension or oscillation.

B. Comparison with real experiment

To assess the simulator's realism, we conducted a real-world experiment with a tethered UAV-UGV system navigating a confined theatre environment (Fig. 1). The UAV and UGV moved actively while the UGV managed the tether, demonstrating the feasibility of planning and executing collision-free trajectories. The experiment was then replicated in Gazebo using a LiDAR reconstruction, maintaining the same parameters, obstacles, and initial conditions.

Two scenarios were designed for comparison. In Real-Time Tether Adaptation (RTTA), the tether length was dynamically adjusted proportionally to the UAV-UGV distance, ensuring real-time slack control. In Predefined Tether Reference (PTR), a fixed tether length was used at each waypoint, mirroring the behaviour of the real-world system, where the tether length is planned as part of the trajectory.

Table II compares the results, showing discrepancies below 0.3% in distance traveled and below 12% in tether length between scenarios. Differences in released and collected tether lengths result from the distinct management strategies in RTTA and PTR. As shown in Fig. 4, both UAV and UGV closely followed their planned paths in simulation and reality. Larger deviations occurred when UAV and UGV were far apart, as the extended tether introduced additional dynamic effects. The tether management system effectively maintained proper slack, preventing excessive tension or looseness that could affect movement.

C. Performance Evaluation

The simulator's computational performance was evaluated for real-time capabilities and scalability by comparing a

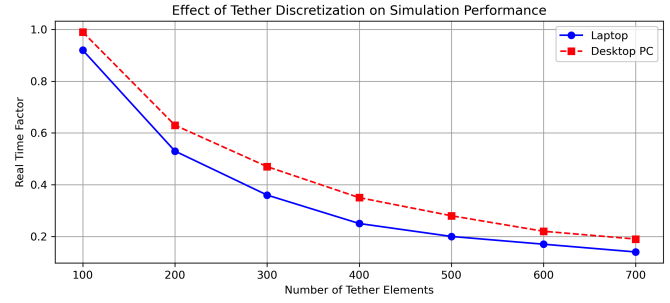


Fig. 5: Performance comparison.

laptop (32 GB RAM, Intel i7-13620H, RTX 4060) and a desktop (64 GB RAM, Intel i9-12900F, RTX 3060). Each test was repeated 10 times, with results averaged due to negligible deviations. All experiments used the ODE Quick-Step solver with a fixed step size of $\Delta t = 4$ ms (250 Hz), 80 solver iterations, SOR relaxation factor 1.0, contact surface layer 1 mm, and CFM = 0.

As shown in Fig. 5, the desktop PC demonstrated superior performance. The number of tether elements directly influenced computational load. For the laptop, the real-time factor (RTF) decreased from 0.92 (100 elements) to 0.14 (700 elements), while the desktop PC maintained higher RTF values, ranging from 0.99 (100 elements) to 0.19 (700 elements), with a tether element length of 0.15 m in all cases. Notably, the desktop achieved an RTF of 0.47 with 300 elements while the laptop required element counts below 200 to maintain $RTF > 0.50$.

These results show that the simulator runs on mid-tier hardware, while higher element counts demand higher-end systems to preserve real-time performance.

V. CONCLUSIONS AND FUTURE WORK

The experiments confirm the simulator's ability to model tethered UAV-UGV dynamics accurately. The UAV experiences minor trajectory disturbances due to complex tether interaction, particularly when UAV and UGV are in close proximity, emphasizing the need for careful tether management.

The simulator effectively replicates real-world trajectories, validating its physical models and control algorithms. This fidelity, together with support for both dynamic and predefined tether length management, makes it a valuable tool for testing algorithms in a risk-free environment before real-world deployment.

Future work will focus on extending the winch model to other ground platforms, integrating joint-force sensing to expose tether tension as a measurable simulation output, and using the framework to characterize the maximum tether-induced disturbance tolerable by different UAV flight controllers. Nonlinear tether dynamics are expected to become relevant under high-speed maneuvers beyond the realistic operating regime of tethered systems; tether-obstacle entanglement has been explored and is illustrated in the supplementary video material, with a more complete treatment left as future work.

SUPPLEMENTARY MATERIALS

This paper is accompanied by supplementary materials providing extended details, additional experimental results, and videos demonstrating key aspects of the simulation framework. The repository contains the full implementation of the simulator, documentation, and extra validation experiments beyond those presented in the paper.

A. GitHub Repository

The GitHub repository¹ contains:

- **Complete source code** of the simulation framework, fully integrated with ROS 2 and Gazebo.
- **Detailed documentation**, including: Model descriptions (UAV, UGV, winch, and tether specifications), Scenario configurations (setup for simulated environments), and Installation instructions (dependencies and Docker-based setup).
- **Additional validation experiments**, including Tether Model Evaluation, which compares the simulated tether's accuracy with a theoretical catenary model. All validation experiments are explained in more detail, covering various aspects of system performance and behavior.
- **ROS 2 launch files** for replicating the experiments presented in the paper.

B. Supplementary Videos

Two videos have been included to illustrate the simulator's capabilities:

- **Video 1 – Simulated and Real Experiment Demonstrations**² This video presents demonstrations of the simulated scenarios used to analyze system behavior. It also includes a direct comparison between simulated and real-world experiments, showcasing how the UAV-UGV system interacts in both settings and validating the fidelity of the simulation.
- **Video 2 – Tether-Obstacle Interactions and Tension Analysis**³ This video focuses on tether-environment interactions, illustrating different collision scenarios between the tether and obstacles. It also provides a qualitative visual assessment of the tether's slack and taut states under different experimental conditions, complementing the quantitative metrics with direct observation of the simulated cable physics.

These supplementary materials offer a comprehensive resource for replicating, modifying, and extending the experiments, ensuring that the simulator can be effectively used for further research in tethered marsupial robotic systems.

¹https://github.com/robotics-upo/marsupial_simulator_ros2

²<https://youtu.be/V7b8kehu9U0>

³<https://youtu.be/2kuzjHIIExs>

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